

ditions, it turns out to be a very critical specification. This number specifies the input leakage current on any I/O when the signal is driven from outside and output leakage when the signal is tristated and the pin is shorted to ground ($V_{IS}=0$). When the signal is configured as an input with a pull-down resistor connected to it, this specification limits the maximum value of the pull-down resistor that can be used. Though the leakage current is expected to decline with increase in the value of the pull-down resistor, the drop across the pull-down resistor may increase substantially and could cross $V_{IL(max)}$ limit.

Glue logic considerations. These days most processors and devices support most of the standard interface controllers, eliminating the need to use external glue logic. Glue logic may still be required for some non-standard/non-typical interfaces, or when on-chip controllers provide a signal protocol that could be used to interface with non-intended devices by using some amount of glue logic. It is recommended to do thorough functional logic and timing analysis for such glue logic interfaces. Additional delays introduced by such glue logic interfaces must be considered carefully. If the interface is bidirectional, delays in both directions must be computed, and this may effectively reduce the maximum operating speed for this interface. Simulate the logic functionality and perform timing analysis before realising any glue logic.

Multifunctional signal pins. Watch out for multifunctional signal pins that have multiplexed signal functionalities. Signal multiplexing schemes are commonly used lately to reduce the package pin count and hence the IC cost and, to some extent, the board design cost.

No product or system design is generic and therefore the likelihood of any system designer wanting to use all signals in any particular design is very rare. So based on careful study and application requirements, signals are multiplexed, and this reduces the effective pin count.

One pin may have multiple func-

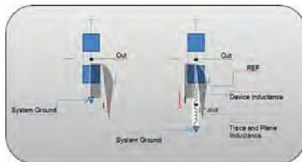


Fig. 5: Ground bounce

tionality such as a GPIO (general purpose I/O), interrupt and memory-select strobe, which can be configured in the software by programming certain control registers. Although the required function could be programmed in the code, upon reset, the pin will be configured by default to have one of the multiple functions. To avoid any signal contention on board, the default configuration during and after reset is normally an input signal (GPIO or an interrupt). The logic circuit should be designed based on what such a multifunctional pin is required to do in a particular system and considering its default state.

Power supply, reset and clock input.

This is a very important specification when designing power supply, chip reset and clock circuitry. Mostly, all processor datasheets begin with this specification. Normally, reset is required to be asserted until all the power supplies are up and stable and for a minimum duration of clock-in cycles. These clock-in cycles are required for the on-chip PLL to lock and function normally. Some initial clock pulses are required to set the chip's internal logic in a known state. Unknown or uninitialised states could cause excess power dissipation during start-up.

When routing, ensure that the high-speed switching signals do not generate inductive or capacitive interference (cross-talk) on signals such as processor clock-in; capacitive interference is more common. This can create problems if the internal PLL design is sensitive to noise or glitches on the clock-in signal.

Unused signals. Never leave unused

signals unattended. Always terminate them appropriately. GPIOs, which are normally configured as inputs by default, when unused can cause excessive power dissipation in the chip. To avoid this situation, these can be configured as outputs in the software.

Signals that have functionalities like external bus request should be deactivated by using external pull-up/down when not used. If not properly deactivated, this could give rise to chip malfunction, lock and software hang condition. Even though certain signals could be later reconfigured in the code to avoid malfunction, there is a time period between the processor's initial code execution like booting and execution of the configuration code. During this period, the system may get affected.

Debug port for prototypes. It is a good idea to have some sort of debug port for the first designed prototypes. A spare UART, if available on the processor, could be used to send out meaningful information to the hyper-terminal of the PC. This would help monitor the health of the system-under-test. In case the software malfunctions, hangs or crashes, this would give an indication as to which region of the code was last executed by the processor before it crashed.

If UART is not available, the unused GPIO (commonly called as 'flags') could be used to send out the binary-coded information. For example, you could drive 16 different messages with four GPIOs. One could have LEDs connected to these GPIOs or they could be simply probed on the scope (cathode ray oscilloscope/logic analyser).

Shared resource system. In multi-processing systems that use two or more processors, external on-board memory is commonly used as a shared resource. This could be used for sharing data for sequential multi-processing or to